

Conditional statements Zero to Hero Questions List

June 4, 2026

1 C Programming Practice Questions – if Statement Only

Rule: Solve all questions using **only the if statement** (no **else**, **switch**, loops, arrays, functions, etc., unless specifically needed).

1.1 Beginner Level

1.1.1 1. Check whether a number is positive.

Hint: Check if `num > 0`.

1.1.2 2. Check whether a number is negative.

Hint: Check if `num < 0`.

1.1.3 3. Check whether a number is zero.

Hint: Compare with 0.

1.1.4 4. Check whether a number is greater than 100.

Hint: Use `>` operator.

1.1.5 5. Check whether a number is less than 50.

Hint: Use `<` operator.

1.1.6 6. Check whether a number is even.

Hint: `num % 2 == 0`.

1.1.7 7. Check whether a number is odd.

Hint: `num % 2 != 0`.

1.1.8 8. Check whether a character is an uppercase letter.

Hint: ASCII range 'A' to 'Z'.

1.1.9 9. Check whether a character is a lowercase letter.

Hint: ASCII range 'a' to 'z'.

1.1.10 10. Check whether a character is a digit.

Hint: ASCII range '0' to '9'.

1.2 Easy Level

1.2.1 11. Check whether a number is divisible by 5.

Hint: `num % 5 == 0`.

1.2.2 12. Check whether a number is divisible by 10.

Hint: Last digit should be 0.

1.2.3 13. Check whether a number is divisible by both 2 and 3.

Hint: Use logical AND.

1.2.4 14. Check whether a number is a multiple of 7.

Hint: Remainder should be 0.

1.2.5 15. Check whether a year is a leap year (basic rule).

Hint: Divisible by 4.

1.2.6 16. Check whether a person is eligible to vote.

Hint: Age $\geq$ 18.

1.2.7 17. Check whether a student has passed.

Hint: Marks $\geq$ passing marks.

1.2.8 18. Check whether temperature is above boiling point.

Hint: Water boils at 100°C.

1.2.9 19. Check whether a number lies between 1 and 100.

Hint: Use two conditions.

1.2.10 20. Check whether a character is a vowel.

Hint: Compare with a, e, i, o, u.

1.3 Intermediate Level

1.3.1 21. Check whether a number is a three-digit number.

Hint: 100–999.

1.3.2 22. Check whether a number is a four-digit number.

Hint: 1000–9999.

1.3.3 23. Check whether a number is divisible by both 5 and 11.

Hint: Use `&&`.

1.3.4 24. Check whether a character is an alphabet.

Hint: Uppercase OR lowercase range.

1.3.5 25. Check whether a character is a special symbol.

Hint: Not alphabet and not digit.

1.3.6 26. Check whether a number is a factor of another number.

Hint: `larger % smaller == 0`.

1.3.7 27. Check whether a number is a multiple of another number.

Hint: Similar to factor checking.

1.3.8 28. Check whether salary exceeds 50,000.

Hint: Compare salary value.

1.3.9 29. Check whether a number is divisible by 9.

Hint: Remainder should be 0.

1.3.10 30. Check whether a number is within a specified range.

Hint: `Lower limit <= num <= upper limit`.

1.4 Advanced Level

1.4.1 31. Check whether a year is a century year.

Hint: Divisible by 100.

1.4.2 32. Check whether a year is divisible by 400.

Hint: Use `% 400`.

1.4.3 33. Check whether a number is a perfect square.

Hint: Square root $\times$ square root = number.

1.4.4 34. Check whether a number is positive and even.

Hint: Combine two conditions.

1.4.5 35. Check whether a number is negative and odd.

Hint: Combine two conditions.

1.4.6 36. Check whether a character is a hexadecimal digit.

Hint: 0–9, A–F, a–f.

1.4.7 37. Check whether a number is within the ASCII range.

Hint: 0–127.

1.4.8 38. Check whether a person is eligible for a senior citizen concession.

Hint: Age $\geq$ 60.

1.4.9 39. Check whether a mark qualifies for distinction.

Hint: Marks $\geq$ 75 (or chosen criterion).

1.4.10 40. Check whether an employee qualifies for a bonus.

Hint: Salary/experience-based condition.

1.5 Challenge Level

1.5.1 41. Check whether a number is a single-digit number.

Hint: -9 to 9.

1.5.2 42. Check whether a number is a two-digit number.

Hint: 10–99.

1.5.3 43. Check whether a number is divisible by 2, 3, and 5 simultaneously.

Hint: Use multiple AND conditions.

1.5.4 44. Check whether a character is a valid C identifier starting character.

Hint: Letter or underscore.

1.5.5 45. Check whether a number belongs to the first quadrant condition.

Hint: Positive x and positive y.

1.5.6 46. Check whether a number is greater than its square root.

Hint: Compare with `sqrt()` result.

1.5.7 47. Check whether a given ASCII value represents a printable character.

Hint: ASCII 32–126.

1.5.8 48. Check whether a character is a consonant.

Hint: Alphabet but not vowel.

1.5.9 49. Check whether a number is divisible by either 4 or 6.

Hint: Use logical OR.

1.5.10 50. Check whether a number satisfies a custom condition:

$x^2 > 100$ **Hint:** Compute square and compare.

1.6 Expert Practice

1.6.1 51. Check whether a point lies on the X-axis.

Hint: `y == 0`.

1.6.2 52. Check whether a point lies on the Y-axis.

Hint: `x == 0`.

1.6.3 53. Check whether a point lies at the origin.

Hint: `x == 0 && y == 0`.

1.6.4 54. Check whether a number is within an absolute limit.

Hint: `abs(num) <= limit`.

1.6.5 55. Check whether a character is a valid binary digit.

Hint: Only 0 or 1.

1.6.6 56. Check whether a number is divisible by its last digit.

Hint: Extract last digit first.

1.6.7 57. Check whether a number is greater than the average of two numbers.

Hint: Calculate average and compare.

1.6.8 58. Check whether a number lies outside a given range.

Hint: Use OR condition.

1.6.9 59. Check whether a user-entered password length meets minimum criteria.

Hint: Check string length.

1.6.10 60. Check whether a triangle can be formed from three sides.

Hint: Triangle inequality theorem.

These 60 problems cover nearly every common use of the **simple if statement** and are excellent preparation before moving to **if-else**, **nested if**, and **else-if ladder** programs.

2 C Programming Practice Questions – if-else Statement Only

Rule: Solve all questions using **only if-else statements**. Avoid **else-if**, **switch**, **loops**, **arrays**, **functions**, etc., unless specifically required.

3 Beginner Level

3.0.1 1. Check whether a number is positive or negative.

Hint: Compare with 0.

3.0.2 2. Check whether a number is even or odd.

Hint: Use % 2.

3.0.3 3. Find the greater of two numbers.

Hint: Compare the two values.

3.0.4 4. Check whether a character is uppercase or lowercase.

Hint: Use ASCII ranges.

3.0.5 5. Check whether a number is divisible by 5 or not.

Hint: Use % 5.

3.0.6 6. Check whether a student passed or failed.

Hint: Compare marks with passing marks.

3.0.7 7. Check whether a person is eligible to vote or not.

Hint: Age $\geq$ 18.

3.0.8 8. Check whether a number is zero or non-zero.

Hint: Compare with 0.

3.0.9 9. Check whether a character is a vowel or consonant.

Hint: Compare against vowels.

3.0.10 10. Check whether a year is a leap year or not (basic rule).

Hint: Divisible by 4.

4 Easy Level

4.0.1 11. Find the smaller of two numbers.

Hint: Compare using $<$.

4.0.2 12. Check whether a number is a multiple of 10.

Hint: Use $\% 10$.

4.0.3 13. Check whether a character is an alphabet or not.

Hint: Check uppercase and lowercase ranges.

4.0.4 14. Check whether a character is a digit or not.

Hint: ASCII range '0' to '9'.

4.0.5 15. Check whether temperature is above freezing point.

Hint: Compare with 0°C .

4.0.6 16. Check whether salary is above 50,000.

Hint: Use comparison operator.

4.0.7 17. Check whether a number is a three-digit number.

Hint: Range 100–999.

4.0.8 18. Check whether a number is divisible by both 2 and 3.

Hint: Use logical AND.

4.0.9 19. Check whether a number is greater than 100.

Hint: Compare directly.

4.0.10 20. Check whether a character is a special symbol.

Hint: Not alphabet and not digit.

5 Intermediate Level

5.0.1 21. Find the larger of two floating-point numbers.

Hint: Use float comparison.

5.0.2 22. Check whether a year is a century year.

Hint: Divisible by 100.

5.0.3 23. Check whether a number is positive or non-positive.

Hint: Use > 0 .

5.0.4 24. Check whether a character is printable or non-printable.

Hint: ASCII 32–126.

5.0.5 25. Check whether a number is a factor of another number.

Hint: Use remainder operator.

5.0.6 26. Check whether a number is within a specified range.

Hint: Use two conditions.

5.0.7 27. Check whether a number is outside a specified range.

Hint: Use logical OR.

5.0.8 28. Check whether a user is eligible for a driving license.

Hint: Age ≥ 18 .

5.0.9 29. Check whether a number is divisible by 7.

Hint: Use $\% 7$.

5.0.10 30. Check whether a character is a hexadecimal digit.

Hint: 0–9, A–F, a–f.

6 Advanced Level

6.0.1 31. Check whether a number is a perfect square.

Hint: Use `sqrt()`.

6.0.2 32. Check whether a point lies on the X-axis.

Hint: `y == 0`.

6.0.3 33. Check whether a point lies on the Y-axis.

Hint: `x == 0`.

6.0.4 34. Check whether a point is at the origin.

Hint: Both coordinates are 0.

6.0.5 35. Check whether a number is positive and even.

Hint: Combine conditions.

6.0.6 36. Check whether a number is negative and odd.

Hint: Combine conditions.

6.0.7 37. Check whether a number is divisible by 11.

Hint: Use `% 11`.

6.0.8 38. Check whether a student obtained distinction.

Hint: Marks > specified value.

6.0.9 39. Check whether a person qualifies for a senior citizen concession.

Hint: Age > 60.

6.0.10 40. Check whether a character is a valid binary digit.

Hint: Only 0 or 1.

7 Challenge Level

7.0.1 41. Compare two numbers and print whether the first is larger.

Hint: Use a simple comparison.

7.0.2 42. Check whether a number is divisible by both 5 and 11.

Hint: Use logical AND.

7.0.3 43. Check whether a number belongs to the interval [50,100].

Hint: Inclusive range.

7.0.4 44. Check whether a character can start a C identifier.

Hint: Letter or underscore.

7.0.5 45. Check whether a number exceeds its square root.

Hint: Compare with `sqrt(num)`.

7.0.6 46. Check whether a given ASCII code is a letter.

Hint: ASCII ranges.

7.0.7 47. Check whether a number is a single-digit number.

Hint: -9 to 9.

7.0.8 48. Check whether a number is a two-digit number.

Hint: 10–99.

7.0.9 49. Check whether a number is divisible by its last digit.

Hint: Extract last digit using `% 10`.

7.0.10 50. Check whether a triangle can be formed from three sides.

Hint: Triangle inequality theorem.

8 Expert Practice

8.0.1 51. Check whether an employee is eligible for a bonus.

Hint: Compare experience or salary.

8.0.2 52. Check whether a customer gets a discount.

Hint: Compare purchase amount.

8.0.3 53. Check whether a number is greater than the average of two numbers.

Hint: Calculate average first.

8.0.4 54. Check whether a number lies inside a circle of radius r .

Hint: Use $x^2 + y^2$.

8.0.5 55. Check whether a number is divisible by 4.

Hint: Use remainder operator.

8.0.6 56. Check whether a given angle is acute or not.

Hint: Less than 90°.

8.0.7 57. Check whether a character is a lowercase letter.

Hint: ASCII range.

8.0.8 58. Check whether a number is an exact multiple of another number.

Hint: Remainder should be zero.

8.0.9 59. Check whether a number is larger than 1000.

Hint: Simple comparison.

8.0.10 60. Check whether a password length satisfies a minimum requirement.

Hint: Use string length (`strlen()`).

8.0.11 61. Check whether a given month number is valid.

Hint: Range 1–12.

8.0.12 62. Check whether a given day number is valid.

Hint: Range 1–31.

8.0.13 63. Check whether a number is divisible by either 4 or 6.

Hint: Use logical OR.

8.0.14 64. Check whether a person can enter a movie (age restriction).

Hint: Minimum age condition.

8.0.15 65. Check whether a product weight exceeds a shipping limit.

Hint: Compare with maximum allowed weight.

8.0.16 66. Check whether electricity consumption exceeds a threshold.

Hint: Compare units consumed.

8.0.17 67. Check whether a score qualifies for a reward.

Hint: Compare score with target.

8.0.18 68. Check whether a number is within machine-safe operating limits.

Hint: Use lower and upper bounds.

8.0.19 69. Check whether an entered character is a punctuation mark.

Hint: ASCII punctuation ranges.

8.0.20 70. Check whether a number satisfies $x^2 > 500$.

Hint: Compute square and compare.

These **70 problems** provide thorough practice of the **simple if-else statement** before moving to **nested if-else** and **else-if ladder** problems.

9 C Programming Practice Questions – if - else if - else Ladder

Rule: Solve all questions using **if - else if - else** statements. Avoid **switch**, loops, arrays, functions, etc., unless specifically required.

10 Beginner Level

10.0.1 1. Find the largest of two numbers.

Hint: Compare the two numbers.

10.0.2 2. Find the largest of three numbers.

Hint: Multiple comparisons are needed.

10.0.3 3. Find the smallest of three numbers.

Hint: Check all possible minimum cases.

10.0.4 4. Check whether a number is positive, negative, or zero.

Hint: Three possible outcomes.

10.0.5 5. Check whether a character is uppercase, lowercase, or a digit.

Hint: Use ASCII ranges.

10.0.6 6. Check whether a number is even, odd, or zero.

Hint: Zero is a separate case.

10.0.7 7. Assign grades based on marks.

Hint: Example: A, B, C, D, F.

10.0.8 8. Classify a person's age group.

Hint: Child, Teenager, Adult, Senior Citizen.

10.0.9 9. Check whether a character is a vowel, consonant, or non-alphabet.

Hint: Use alphabet checks first.

10.0.10 10. Determine profit, loss, or no profit/no loss.

Hint: Compare cost price and selling price.

11 Easy Level

11.0.1 11. Find the largest among four numbers.

Hint: Multiple conditions.

11.0.2 12. Check whether a year is leap year, century year, or normal year.

Hint: Use divisibility rules.

11.0.3 13. Classify temperature.

Hint: Freezing, Cold, Warm, Hot.

11.0.4 14. Determine the quadrant of a point (x, y).

Hint: Four quadrants plus axes.

11.0.5 15. Check whether a character is a letter, digit, or special character.

Hint: ASCII classification.

11.0.6 16. Classify a student's percentage.

Hint: Distinction, First Class, Second Class, Pass, Fail.

11.0.7 17. Determine ticket pricing based on age.

Hint: Different age categories.

11.0.8 18. Calculate bonus percentage based on years of service.

Hint: Multiple service slabs.

11.0.9 19. Determine electricity bill rate slab.

Hint: Different unit ranges.

11.0.10 20. Categorize BMI.

Hint: Underweight, Normal, Overweight, Obese.

12 Intermediate Level

12.0.1 21. Find the middle number among three numbers.

Hint: Neither largest nor smallest.

12.0.2 22. Determine the type of triangle based on sides.

Hint: Equilateral, Isosceles, Scalene.

12.0.3 23. Determine the type of angle.

Hint: Acute, Right, Obtuse, Straight.

12.0.4 24. Determine the season based on month number.

Hint: Group months into seasons.

12.0.5 25. Determine the number of days in a month.

Hint: February, 30-day months, 31-day months.

12.0.6 26. Check whether a character is uppercase, lowercase, digit, whitespace, or special character.

Hint: Multiple ASCII ranges.

12.0.7 27. Categorize income tax slab.

Hint: Different income ranges.

12.0.8 28. Determine scholarship eligibility.

Hint: Multiple percentage bands.

12.0.9 29. Determine shipping charges based on package weight.

Hint: Use weight slabs.

12.0.10 30. Categorize exam performance.

Hint: Excellent, Good, Average, Poor.

13 Advanced Level

13.0.1 31. Solve a simple calculator operation.

Hint: User enters operator (+, -, *, /).

13.0.2 32. Determine zodiac sign based on month and date.

Hint: Use date ranges.

13.0.3 33. Determine blood pressure category.

Hint: Multiple ranges.

13.0.4 34. Categorize internet speed.

Hint: Slow, Medium, Fast, Very Fast.

13.0.5 35. Determine loan eligibility category.

Hint: Based on credit score.

13.0.6 36. Determine employee designation based on experience.

Hint: Different experience ranges.

13.0.7 37. Determine voting eligibility category.

Hint: Minor, Adult, Senior.

13.0.8 38. Classify rainfall intensity.

Hint: Light, Moderate, Heavy.

13.0.9 39. Categorize fuel efficiency.

Hint: Excellent, Good, Average, Poor.

13.0.10 40. Determine movie ticket category based on seat class.

Hint: Silver, Gold, Platinum.

14 Challenge Level

14.0.1 41. Find the largest among five numbers.

Hint: Compare all possibilities.

14.0.2 42. Determine the type of quadrilateral.

Hint: Based on side lengths and angles.

14.0.3 43. Categorize a student's CGPA.

Hint: Multiple CGPA ranges.

14.0.4 44. Determine game level based on score.

Hint: Beginner, Intermediate, Advanced, Expert.

14.0.5 45. Classify a number based on digit count.

Hint: Single, Double, Triple, Multi-digit.

14.0.6 46. Determine discount percentage based on purchase amount.

Hint: Different purchase slabs.

14.0.7 47. Categorize battery percentage.

Hint: Critical, Low, Medium, Full.

14.0.8 48. Determine weather alert level.

Hint: Green, Yellow, Orange, Red.

14.0.9 49. Categorize internet data usage.

Hint: Low, Moderate, High.

14.0.10 50. Determine customer membership tier.

Hint: Bronze, Silver, Gold, Platinum.

15 Expert Level

15.0.1 51. Determine the day of the week from a number (1–7).

Hint: Use ladder conditions.

15.0.2 52. Determine month name from month number.

Hint: 1–12 mapping.

15.0.3 53. Determine academic standing based on GPA.

Hint: Several GPA ranges.

15.0.4 54. Categorize heart rate.

Hint: Low, Normal, Elevated, High.

15.0.5 55. Determine risk category based on age.

Hint: Multiple age groups.

15.0.6 56. Classify vehicle speed.

Hint: Slow, Normal, Fast, Overspeed.

15.0.7 57. Determine salary grade.

Hint: Salary slabs.

15.0.8 58. Categorize air quality index (AQI).

Hint: Good, Moderate, Poor, Severe.

15.0.9 59. Determine credit rating.

Hint: Excellent, Good, Fair, Poor.

15.0.10 60. Classify water consumption.

Hint: Low, Average, High.

16 Real-World Scenario Problems

16.0.1 61. Mobile recharge plan selection.

Hint: Different price ranges.

16.0.2 62. Internet package recommendation.

Hint: Based on usage.

16.0.3 63. Hotel room category assignment.

Hint: Budget, Standard, Deluxe, Suite.

16.0.4 64. Employee performance appraisal.

Hint: Outstanding, Good, Average, Poor.

16.0.5 65. ATM withdrawal charge calculation.

Hint: Amount-based slabs.

16.0.6 66. Parking fee determination.

Hint: Time-based slabs.

16.0.7 67. Courier service category.

Hint: Weight ranges.

16.0.8 68. Library fine calculation category.

Hint: Delay ranges.

16.0.9 69. Insurance premium category.

Hint: Age groups.

16.0.10 70. Admission eligibility category.

Hint: Percentage cutoffs.

17 Master-Level Practice

17.0.1 71. Menu-driven calculator using if-else-if.

Hint: Option numbers 1–4.

17.0.2 72. Determine rank from marks.

Hint: Topper, First, Second, Third.

17.0.3 73. Categorize machine temperature.

Hint: Safe, Warning, Critical.

17.0.4 74. Categorize server load.

Hint: Low, Medium, High.

17.0.5 75. Determine fuel level status.

Hint: Empty, Low, Medium, Full.

17.0.6 76. Categorize stock availability.

Hint: Out of Stock, Low Stock, Available.

17.0.7 77. Determine customer satisfaction rating.

Hint: Excellent to Poor.

17.0.8 78. Categorize response time.

Hint: Fast, Moderate, Slow.

17.0.9 79. Determine network signal strength.

Hint: Excellent, Good, Weak.

17.0.10 80. Categorize project completion percentage.

Hint: Not Started, In Progress, Nearly Done, Completed.

These **80 problems** cover almost every pattern of **if → else if → else** ladder programming that commonly appears in C programming courses, lab exams, interviews, and beginner competitive programming.

18 C Programming Practice Questions – Nested if / Nested if-else

Rule: Solve all questions using **nested if** or **nested if-else** statements only. Avoid **else-if** ladders unless absolutely necessary.

A nested **if** means an **if** statement inside another **if** statement.

19 Beginner Level

19.0.1 1. Check whether a number is positive and even.

Hint: First check positivity, then check evenness.

19.0.2 2. Check whether a number is positive and odd.

Hint: Second condition is checked only if first is true.

19.0.3 3. Check whether a person is eligible to vote and has a voter ID.

Hint: Age check inside another condition.

19.0.4 4. Check whether a student passed and obtained distinction.

Hint: Pass condition first.

19.0.5 5. Check whether a number is divisible by 2 and 3.

Hint: Nested divisibility checks.

19.0.6 6. Check whether a character is an alphabet and then a vowel.

Hint: Alphabet check first.

19.0.7 7. Check whether a year is divisible by 4 and then by 100.

Hint: Nested leap-year logic.

19.0.8 8. Check whether a number is greater than 0 and less than 100.

Hint: Range validation using nesting.

19.0.9 9. Check whether a salary qualifies for a bonus and then an additional incentive.

Hint: Bonus eligibility first.

19.0.10 10. Check whether a customer is a member and eligible for a discount.

Hint: Membership condition first.

20 Easy Level

20.0.1 11. Find the largest of two positive numbers.

Hint: Verify positivity before comparison.

20.0.2 12. Check whether a character is uppercase and a vowel.

Hint: Uppercase check first.

20.0.3 13. Check whether a number is divisible by 5 and 11.

Hint: Two-level divisibility test.

20.0.4 14. Check whether a student passed all subjects and achieved first class.

Hint: Subject pass condition first.

20.0.5 15. Check whether a triangle can be formed and then classify it as equilateral.

Hint: Triangle validity first.

20.0.6 16. Check whether a number is a three-digit number and even.

Hint: Digit count first.

20.0.7 17. Check whether a password length is valid and contains a digit.

Hint: Length validation first.

20.0.8 18. Check whether a person is eligible for a driving license and owns required documents.

Hint: Age check first.

20.0.9 19. Check whether a product is in stock and qualifies for free shipping.

Hint: Stock status first.

20.0.10 20. Check whether a character is a lowercase alphabet and consonant.

Hint: Alphabet check before vowel test.

21 Intermediate Level

21.0.1 21. Find the largest of three numbers using nested if.

Hint: Compare first two, then compare winner with third.

21.0.2 22. Find the smallest of three numbers using nested if.

Hint: Similar logic as largest.

21.0.3 23. Determine whether a year is a leap year using nested conditions.

Hint: Divisible by 4 $\rightarrow$ 100 $\rightarrow$ 400.

21.0.4 24. Check whether a number is positive, even, and divisible by 5.

Hint: Three nested levels.

21.0.5 25. Determine whether a point lies in the first quadrant.

Hint: Check x first, then y.

21.0.6 26. Check whether a character is an alphabet and then classify as uppercase or lowercase.

Hint: Nested ASCII checks.

21.0.7 27. Check whether a number is a factor of another number and also even.

Hint: Factor condition first.

21.0.8 28. Check whether a candidate is eligible for a job and has required experience.

Hint: Qualification first.

21.0.9 29. Check whether a student passed and qualifies for a scholarship.

Hint: Pass percentage first.

21.0.10 30. Determine whether a number is a perfect square and even.

Hint: Square test first.

22 Advanced Level

22.0.1 31. Determine whether a triangle is valid and classify as equilateral, isosceles, or scalene.

Hint: Validity check first.

22.0.2 32. Check whether a point lies inside a circle and in the first quadrant.

Hint: Circle condition first.

22.0.3 33. Check whether a number is positive and a multiple of both 3 and 7.

Hint: Multiple nested conditions.

22.0.4 34. Determine employee promotion eligibility.

Hint: Experience $\rightarrow$ Performance $\rightarrow$ Promotion.

22.0.5 35. Determine admission eligibility.

Hint: Marks $\rightarrow$ Entrance Score.

22.0.6 36. Check whether a character is printable and alphabetic.

Hint: Printable range first.

22.0.7 37. Determine whether a customer qualifies for premium membership.

Hint: Spending threshold $\rightarrow$ Membership duration.

22.0.8 38. Check whether a machine is running within safe temperature and pressure limits.

Hint: Temperature first.

22.0.9 39. Check whether a number is within a range and divisible by a given number.

Hint: Range validation first.

22.0.10 40. Determine whether a bank account qualifies for special interest rates.

Hint: Balance $\rightarrow$ Account age.

23 Challenge Level

23.0.1 41. Find the largest of four numbers using nested if.

Hint: Compare progressively.

23.0.2 42. Find the second largest among three numbers.

Hint: Nested comparison tree.

23.0.3 43. Determine whether a student is eligible for placement.

Hint: Percentage → Backlogs → Skills test.

23.0.4 44. Check whether a user can access an admin panel.

Hint: Login → Role verification.

23.0.5 45. Determine whether a loan applicant is eligible.

Hint: Income → Credit score → Employment status.

23.0.6 46. Check whether a vehicle is eligible for emission certification.

Hint: Vehicle age → Emission test.

23.0.7 47. Determine whether a file can be uploaded.

Hint: File size → File type.

23.0.8 48. Check whether a number is prime and greater than 100.

Hint: Prime check first.

23.0.9 49. Determine whether an internet connection qualifies for a premium plan.

Hint: Speed → Usage.

23.0.10 50. Check whether a user qualifies for a loyalty reward.

Hint: Membership → Purchase amount.

24 Expert Level

24.0.1 51. Determine employee bonus level.

Hint: Experience → Salary → Rating.

24.0.2 52. Determine scholarship category.

Hint: Marks → Income → Category.

24.0.3 53. Check whether a customer gets a discount and cashback.

Hint: Purchase amount → Payment mode.

24.0.4 54. Determine insurance eligibility.

Hint: Age → Health score.

24.0.5 55. Check whether a student can register for an advanced course.

Hint: Prerequisite course → Grade.

24.0.6 56. Determine whether a project qualifies for funding.

Hint: Budget → Feasibility → Approval.

24.0.7 57. Check whether a passenger can board an international flight.

Hint: Ticket → Passport → Visa.

24.0.8 58. Determine whether a machine can start.

Hint: Power → Safety switch → Temperature.

24.0.9 59. Check whether a warehouse item can be dispatched.

Hint: Stock → Payment → Packaging.

24.0.10 60. Determine whether a user qualifies for VIP status.

Hint: Membership → Purchases → Activity.

25 Real-World Nested if Practice

25.0.1 61. ATM withdrawal validation.

Hint: PIN → Balance → Withdrawal limit.

25.0.2 62. Online exam access.

Hint: Registration → Fee payment → Hall ticket.

25.0.3 63. E-commerce order processing.

Hint: Stock → Payment → Shipping.

25.0.4 64. Hostel admission eligibility.

Hint: Admission confirmed → Room availability.

25.0.5 65. Hospital appointment booking.

Hint: Registration → Doctor availability.

25.0.6 66. Library book issue system.

Hint: Membership → Fine status → Book availability.

25.0.7 67. Employee attendance bonus.

Hint: Attendance → Performance.

25.0.8 68. Mobile recharge offer eligibility.

Hint: Plan type → Recharge amount.

25.0.9 69. Sports team selection.

Hint: Fitness → Skill score.

25.0.10 70. Internship eligibility.

Hint: CGPA → Skills → Interview score.

26 Master-Level Nested if Problems

26.0.1 71. Find the largest of five numbers using only nested if.

Hint: Build a comparison hierarchy.

26.0.2 72. Determine election candidacy eligibility.

Hint: Age → Citizenship → Criminal record.

26.0.3 73. Validate a strong password.

Hint: Length → Digit → Special character.

26.0.4 74. Determine premium credit card eligibility.

Hint: Income → Credit score → Existing debts.

26.0.5 75. University admission process.

Hint: Marks → Entrance Exam → Interview.

26.0.6 76. Vehicle registration approval.

Hint: Documents → Insurance → Fitness certificate.

26.0.7 77. Loan approval workflow.

Hint: Income → Credit score → Collateral.

26.0.8 78. Industrial machine startup sequence.

Hint: Power → Pressure → Temperature → Safety lock.

26.0.9 79. Airport security clearance.

Hint: Ticket → ID → Security check.

26.0.10 80. Multi-level user authentication.

Hint: Username → Password → OTP → Security Question.

These **80 problems** focus specifically on building logic with **nested if** and **nested if-else statements**, which is the next major step after mastering simple **if**, **if-else**, and **if-else-if** structures in C.

27 C Programming Practice Questions – else-if Ladder Only

Rule: Solve these problems primarily using an **if → else if → else** ladder. The goal is to choose **one outcome from many possible conditions**.

28 Beginner Level

28.0.1 1. Determine whether a number is positive, negative, or zero.

Hint: Three possible outcomes.

28.0.2 2. Assign grades based on marks.

Hint: A, B, C, D, F.

28.0.3 3. Classify a person's age group.

Hint: Child, Teenager, Adult, Senior Citizen.

28.0.4 4. Determine whether a temperature is Cold, Warm, or Hot.

Hint: Use temperature ranges.

28.0.5 5. Categorize a student's percentage.

Hint: Distinction, First Class, Second Class, Pass, Fail.

28.0.6 6. Classify BMI.

Hint: Underweight, Normal, Overweight, Obese.

28.0.7 7. Determine a salary category.

Hint: Low, Medium, High Income.

28.0.8 8. Categorize rainfall intensity.

Hint: Light, Moderate, Heavy.

28.0.9 9. Determine fuel level status.

Hint: Empty, Low, Medium, Full.

28.0.10 10. Categorize battery percentage.

Hint: Critical, Low, Medium, High.

29 Easy Level

29.0.1 11. Determine tax slab based on annual income.

Hint: Multiple income ranges.

29.0.2 12. Categorize internet speed.

Hint: Slow, Moderate, Fast, Very Fast.

29.0.3 13. Determine scholarship amount based on marks.

Hint: Different percentage bands.

29.0.4 14. Classify a student's CGPA.

Hint: Excellent, Good, Average, Poor.

29.0.5 15. Determine an employee's designation based on experience.

Hint: Fresher, Junior, Senior, Lead.

29.0.6 16. Categorize electricity consumption.

Hint: Low, Average, High.

29.0.7 17. Determine shipping charges based on package weight.

Hint: Weight slabs.

29.0.8 18. Categorize water consumption.

Hint: Low, Moderate, High.

29.0.9 19. Determine mobile recharge benefits.

Hint: Recharge amount ranges.

29.0.10 20. Classify response time.

Hint: Fast, Acceptable, Slow.

30 Intermediate Level

30.0.1 21. Determine a loan interest rate based on credit score.

Hint: Score ranges.

30.0.2 22. Categorize employee performance.

Hint: Outstanding, Good, Average, Poor.

30.0.3 23. Determine vehicle speed category.

Hint: Slow, Normal, Fast, Overspeed.

30.0.4 24. Classify air quality index (AQI).

Hint: Good, Moderate, Poor, Severe.

30.0.5 25. Determine parking fee slab.

Hint: Parking duration ranges.

30.0.6 26. Categorize machine efficiency.

Hint: Excellent, Good, Average, Poor.

30.0.7 27. Determine hotel room category.

Hint: Budget, Standard, Deluxe, Suite.

30.0.8 28. Categorize memory usage.

Hint: Normal, Warning, Critical.

30.0.9 29. Determine insurance premium category.

Hint: Age-based groups.

30.0.10 30. Classify customer membership.

Hint: Bronze, Silver, Gold, Platinum.

31 Advanced Level

31.0.1 31. Determine the quadrant of a point.

Hint: Use x and y values.

31.0.2 32. Categorize exam performance.

Hint: Excellent to Fail.

31.0.3 33. Determine weather alert level.

Hint: Green, Yellow, Orange, Red.

31.0.4 34. Categorize production quality.

Hint: Grade A, B, C, Reject.

31.0.5 35. Determine discount percentage based on purchase amount.

Hint: Purchase slabs.

31.0.6 36. Categorize project completion percentage.

Hint: Not Started, In Progress, Near Completion, Completed.

31.0.7 37. Determine server load status.

Hint: Low, Medium, High, Critical.

31.0.8 38. Categorize internet data usage.

Hint: Low, Medium, High.

31.0.9 39. Determine network signal strength.

Hint: Excellent, Good, Weak, Poor.

31.0.10 40. Categorize stock inventory level.

Hint: Out of Stock, Low, Medium, High.

32 Challenge Level

32.0.1 41. Determine university admission status.

Hint: Admission, Waiting List, Rejected.

32.0.2 42. Categorize industrial machine temperature.

Hint: Safe, Warning, Danger.

32.0.3 43. Determine software license status.

Hint: Active, Expiring Soon, Expired.

32.0.4 44. Categorize hospital patient priority.

Hint: Low, Medium, High, Emergency.

32.0.5 45. Determine flight fare category.

Hint: Economy, Premium Economy, Business, First Class.

32.0.6 46. Categorize customer satisfaction score.

Hint: Excellent, Good, Average, Poor.

32.0.7 47. Determine maintenance urgency.

Hint: Routine, Important, Critical.

32.0.8 48. Categorize network latency.

Hint: Excellent, Good, Fair, Poor.

32.0.9 49. Determine risk level for an investment.

Hint: Low, Medium, High Risk.

32.0.10 50. Categorize CPU utilization.

Hint: Idle, Normal, Busy, Overloaded.

33 Expert Level

33.0.1 51. Determine employee salary grade.

Hint: Grade A–E based on salary.

33.0.2 52. Categorize annual company performance.

Hint: Excellent, Good, Average, Poor.

33.0.3 53. Determine smart irrigation status.

Hint: Dry, Moderate, Wet.

33.0.4 54. Categorize traffic congestion.

Hint: Light, Moderate, Heavy, Severe.

33.0.5 55. Determine internet plan recommendation.

Hint: Based on monthly usage.

33.0.6 56. Categorize healthcare risk.

Hint: Low, Moderate, High.

33.0.7 57. Determine loan approval category.

Hint: Approved, Conditional, Rejected.

33.0.8 58. Categorize factory production output.

Hint: Low, Medium, High.

33.0.9 59. Determine student rank category.

Hint: Topper, Excellent, Good, Average.

33.0.10 60. Categorize operating system memory availability.

Hint: Healthy, Warning, Critical.

34 Real-World Scenario Problems

34.0.1 61. Mobile battery health status.

Hint: Percentage ranges.

34.0.2 62. Water tank level monitoring.

Hint: Empty, Low, Medium, Full.

34.0.3 63. Fuel station warning system.

Hint: Fuel percentage ranges.

34.0.4 64. Smart AC cooling mode.

Hint: Temperature bands.

34.0.5 65. ATM cash availability status.

Hint: Low, Medium, High cash.

34.0.6 66. Industrial pressure monitoring.

Hint: Safe, Warning, Critical.

34.0.7 67. Student attendance category.

Hint: Excellent, Good, Shortage.

34.0.8 68. Employee attendance rating.

Hint: Attendance percentage ranges.

34.0.9 69. Warehouse stock classification.

Hint: Stock ranges.

34.0.10 70. E-commerce customer loyalty level.

Hint: Purchase history ranges.

35 Master-Level Practice

35.0.1 71. Classify machine vibration levels.

Hint: Safe, Warning, Dangerous.

35.0.2 72. Determine cybersecurity threat level.

Hint: Low, Medium, High, Critical.

35.0.3 73. Categorize cloud storage utilization.

Hint: Free, Moderate, High, Full.

35.0.4 74. Determine academic standing.

Hint: Honors, Good Standing, Probation.

35.0.5 75. Categorize internet bandwidth usage.

Hint: Low, Moderate, High.

35.0.6 76. Determine energy consumption category.

Hint: Low, Medium, High.

35.0.7 77. Classify hospital bed occupancy.

Hint: Available, Busy, Full.

35.0.8 78. Categorize sensor readings.

Hint: Normal, Warning, Critical.

35.0.9 79. Determine project risk category.

Hint: Low, Medium, High.

35.0.10 80. Categorize overall system health.

Hint: Healthy, Stable, Warning, Critical.

These **80 problems** focus specifically on situations where an **else-if ladder** is the most natural and efficient decision-making structure in C. They are ideal practice before learning more advanced combinations involving nested conditions and switch-case menus.

36 C Programming Practice Questions – switch, case, break (and continue where applicable)

Rule: Use **switch-case** wherever possible. Use **break** to prevent fall-through. Use **continue** only in questions that involve a loop along with **switch**.

37 Beginner Level

37.0.1 1. Create a simple calculator.

Hint: Switch on operator (+, -, *, /, %).

37.0.2 2. Print the day name for a given day number (1–7).

Hint: One case for each day.

37.0.3 3. Print the month name for a given month number (1–12).

Hint: 12 cases.

37.0.4 4. Check whether a character is a vowel.

Hint: Cases for a, e, i, o, u.

37.0.5 5. Display the number of days in a month.

Hint: Group months with 31 days and 30 days.

37.0.6 6. Print a menu option selected by the user.

Hint: Cases 1–5.

37.0.7 7. Convert a digit (0–9) into words.

Hint: One case per digit.

37.0.8 8. Print the corresponding grade description.

Hint: Cases for A, B, C, D, F.

37.0.9 9. Display the season based on a month number.

Hint: Multiple months can share a case block.

37.0.10 10. Print the name of a traffic signal color.

Hint: Cases for Red, Yellow, Green.

38 Easy Level

38.0.1 11. Find whether a character is a vowel or consonant.

Hint: Vowels in cases; default for consonants.

38.0.2 12. Build a basic ATM menu.

Hint: Balance, Deposit, Withdraw, Exit.

38.0.3 13. Display department names from department codes.

Hint: Switch on integer code.

38.0.4 14. Convert Roman numerals (I, V, X, L, C) to values.

Hint: Character-based switch.

38.0.5 15. Print weekday or weekend.

Hint: Group cases.

38.0.6 16. Display different ticket prices based on seat category.

Hint: Silver, Gold, Platinum.

38.0.7 17. Convert lowercase vowels to uppercase vowels.

Hint: Switch on character.

38.0.8 18. Create a restaurant menu system.

Hint: Switch on menu choice.

38.0.9 19. Display different discount percentages.

Hint: Switch on membership type.

38.0.10 20. Print the corresponding arithmetic operation name.

Hint: Cases for operators.

39 Intermediate Level

39.0.1 21. Build a menu-driven unit converter.

Hint: Length, Weight, Temperature.

39.0.2 22. Display the number of days in a month considering month number.

Hint: Use grouped cases.

39.0.3 23. Convert grades to GPA points.

Hint: A→4, B→3, etc.

39.0.4 24. Create a language selection menu.

Hint: English, Hindi, Telugu, Tamil.

39.0.5 25. Simulate a vending machine.

Hint: Product code as switch expression.

39.0.6 26. Determine quarter of the year from month number.

Hint: Group months.

39.0.7 27. Display employee designation from code.

Hint: Use integer codes.

39.0.8 28. Build a simple banking menu.

Hint: Deposit, Withdraw, Transfer.

39.0.9 29. Determine a mathematical function to apply.

Hint: Square, Cube, Square Root.

39.0.10 30. Display internet plan details based on plan code.

Hint: Multiple plans.

40 Advanced Level

40.0.1 31. Menu-driven geometry calculator.

Hint: Area of Circle, Rectangle, Triangle.

40.0.2 32. Display CPU scheduling algorithm names from choice number.

Hint: FCFS, SJF, RR, Priority.

40.0.3 33. Simulate a basic operating system command menu.

Hint: Open, Save, Exit.

40.0.4 34. Determine vehicle type from code.

Hint: Bike, Car, Bus, Truck.

40.0.5 35. Create a billing menu for a grocery store.

Hint: Product categories.

40.0.6 36. Convert marks range into grade categories.

Hint: Use marks/10 in switch.

40.0.7 37. Create a file operation menu.

Hint: Create, Delete, Rename.

40.0.8 38. Simulate a remote control menu.

Hint: Volume, Channel, Power.

40.0.9 39. Create a login role menu.

Hint: Admin, Faculty, Student.

40.0.10 40. Build a hotel room booking menu.

Hint: Standard, Deluxe, Suite.

41 Challenge Level

41.0.1 41. Build a menu-driven scientific calculator.

Hint: Trigonometric and logarithmic operations.

41.0.2 42. Convert month number to quarter and season.

Hint: Nested logic inside cases.

41.0.3 43. Create a railway reservation menu.

Hint: Book, Cancel, Status.

41.0.4 44. Simulate a library management menu.

Hint: Issue, Return, Search.

41.0.5 45. Create a hospital management menu.

Hint: Appointment, Billing, Records.

41.0.6 46. Display programming language details from code.

Hint: C, C++, Java, Python.

41.0.7 47. Build a movie ticket booking menu.

Hint: Different screen categories.

41.0.8 48. Create an online shopping category menu.

Hint: Electronics, Clothing, Books.

41.0.9 49. Display tax slab information.

Hint: Switch on slab code.

41.0.10 50. Create a mobile recharge menu.

Hint: Data, Voice, Combo plans.

42 continue + switch Practice (Requires Loops)

42.0.1 51. Read numbers repeatedly; skip processing when user enters 0.

Hint: Use continue.

42.0.2 52. Menu-driven calculator inside a loop.

Hint: Continue after each operation.

42.0.3 53. Process grades repeatedly; skip invalid grades.

Hint: Default case + continue.

42.0.4 54. Employee menu system.

Hint: Continue after displaying details.

42.0.5 55. Student result processing.

Hint: Skip absent students.

42.0.6 56. Vending machine simulation.

Hint: Continue after each purchase.

42.0.7 57. Traffic signal simulator.

Hint: Continue until Exit option selected.

42.0.8 58. ATM simulation loop.

Hint: Continue after each transaction.

42.0.9 59. Quiz application.

Hint: Skip unanswered questions using continue.

42.0.10 60. Inventory management menu.

Hint: Continue processing until Exit.

43 Expert Level

43.0.1 61. Menu-driven student management system.

Hint: Add, Search, Delete, Exit.

43.0.2 62. Menu-driven employee payroll system.

Hint: Salary, Bonus, Tax.

43.0.3 63. University course registration menu.

Hint: Register, Drop, View.

43.0.4 64. Airline reservation system menu.

Hint: Book, Cancel, Check Status.

43.0.5 65. Hospital patient management menu.

Hint: Admission, Discharge, Billing.

43.0.6 66. Banking application menu.

Hint: Multiple transactions.

43.0.7 67. Hotel management system menu.

Hint: Booking and checkout.

43.0.8 68. Parking management menu.

Hint: Entry, Exit, Fee.

43.0.9 69. Inventory control application.

Hint: Add stock, Remove stock, View stock.

43.0.10 70. Multi-level application menu using nested switches.

Hint: Main Menu → Sub Menu → Action.

43.1 Key Concepts Covered

Concept	Questions
Basic switch-case	1–20
Grouped Cases	5, 15, 22, 26
Character-based Switch	4, 11, 14, 17
Menu-driven Programs	18–70
Nested Switch	42, 70
break Usage	Almost all
continue Usage	51–60
Real-world Applications	41–70

These **70 practice problems** cover nearly all common uses of **switch**, **case**, **break**, **nested switch**, and **continue** (with loops) in C programming.

44 C Programming – Conditional Statements (Real-Life Scenario Problems)

Goal: Practice **if**, **if-else**, **else-if**, **nested if**, and **switch** through real-world situations.

45 Education & Student Management

45.0.1 1. Determine whether a student passes or fails.

Hint: Compare marks with passing criteria.

45.0.2 2. Assign grades (A, B, C, D, F).

Hint: Use percentage ranges.

45.0.3 3. Check scholarship eligibility.

Hint: Percentage and family income.

45.0.4 4. Determine distinction, first class, second class, or fail.

Hint: Use mark intervals.

45.0.5 5. Check exam eligibility.

Hint: Attendance percentage requirement.

45.0.6 6. Determine hostel eligibility.

Hint: Distance from home and admission status.

45.0.7 7. Check whether a student can sit for placement interviews.

Hint: Minimum CGPA and no backlogs.

45.0.8 8. Determine tuition fee concession.

Hint: Income-based slabs.

45.0.9 9. Check eligibility for a merit certificate.

Hint: Top score threshold.

45.0.10 10. Determine academic probation status.

Hint: CGPA below a limit.

46 Banking & Finance

46.0.1 11. Check loan eligibility.

Hint: Income and credit score.

46.0.2 12. Determine loan interest rate slab.

Hint: Based on credit score.

46.0.3 13. Check ATM withdrawal approval.

Hint: Balance and withdrawal limit.

46.0.4 14. Determine credit card eligibility.

Hint: Income and employment status.

46.0.5 15. Check whether a transaction should be flagged.

Hint: Amount exceeds threshold.

46.0.6 16. Determine fixed deposit interest category.

Hint: Age and deposit amount.

46.0.7 17. Check overdraft eligibility.

Hint: Account type and balance history.

46.0.8 18. Determine cashback eligibility.

Hint: Purchase amount and payment method.

46.0.9 19. Check minimum balance penalty.

Hint: Compare balance with required minimum.

46.0.10 20. Determine tax slab.

Hint: Income ranges.

47 Shopping & E-Commerce

47.0.1 21. Apply discount based on purchase amount.

Hint: Different spending slabs.

47.0.2 22. Determine free shipping eligibility.

Hint: Minimum order value.

47.0.3 23. Check coupon validity.

Hint: Expiry date and order amount.

47.0.4 24. Determine membership tier.

Hint: Total purchases.

47.0.5 25. Check return eligibility.

Hint: Days since purchase.

47.0.6 26. Determine cashback percentage.

Hint: Membership type.

47.0.7 27. Check stock availability.

Hint: Quantity > 0.

47.0.8 28. Determine delivery priority.

Hint: Premium customer status.

47.0.9 29. Check exchange eligibility.

Hint: Product condition and time limit.

47.0.10 30. Determine reward points earned.

Hint: Purchase amount slabs.

48 Employee & HR Management

48.0.1 31. Determine bonus eligibility.

Hint: Experience and performance rating.

48.0.2 32. Check promotion eligibility.

Hint: Experience and appraisal score.

48.0.3 33. Determine overtime payment.

Hint: Hours worked beyond standard.

48.0.4 34. Check attendance bonus eligibility.

Hint: Attendance percentage.

48.0.5 35. Determine leave approval.

Hint: Available leave balance.

48.0.6 36. Check retirement eligibility.

Hint: Age requirement.

48.0.7 37. Determine salary grade.

Hint: Salary ranges.

48.0.8 38. Check training program eligibility.

Hint: Department and experience.

48.0.9 39. Determine employee category.

Hint: Years of service.

48.0.10 40. Check reimbursement approval.

Hint: Expense amount and policy limit.

49 Hospital & Healthcare

49.0.1 41. Determine patient priority.

Hint: Severity level.

49.0.2 42. Check health insurance eligibility.

Hint: Age and policy status.

49.0.3 43. Determine BMI category.

Hint: BMI ranges.

49.0.4 44. Check blood donation eligibility.

Hint: Age and weight.

49.0.5 45. Determine ICU admission requirement.

Hint: Critical condition indicators.

49.0.6 46. Check vaccination eligibility.

Hint: Age group.

49.0.7 47. Determine consultation fee.

Hint: Patient category.

49.0.8 48. Check surgery eligibility.

Hint: Medical clearance.

49.0.9 49. Determine emergency level.

Hint: Symptoms and vital signs.

49.0.10 50. Check medicine discount eligibility.

Hint: Senior citizen status.

50 Transportation & Travel

50.0.1 51. Determine driving license eligibility.

Hint: Minimum age.

50.0.2 52. Calculate traffic fine category.

Hint: Speed ranges.

50.0.3 53. Check vehicle insurance validity.

Hint: Expiry date.

50.0.4 54. Determine toll fee category.

Hint: Vehicle type.

50.0.5 55. Check railway ticket concession eligibility.

Hint: Age and category.

50.0.6 56. Determine flight boarding eligibility.

Hint: Ticket and ID verification.

50.0.7 57. Check parking fee slab.

Hint: Parking duration.

50.0.8 58. Determine luggage charge.

Hint: Weight ranges.

50.0.9 59. Check emission certificate requirement.

Hint: Vehicle age.

50.0.10 60. Determine travel class upgrade eligibility.

Hint: Membership level.

51 IT & Software Systems

51.0.1 61. Validate login credentials.

Hint: Username and password.

51.0.2 62. Check password strength.

Hint: Length and complexity.

51.0.3 63. Determine user role access.

Hint: Admin, User, Guest.

51.0.4 64. Check software license validity.

Hint: Expiry date.

51.0.5 65. Determine server load status.

Hint: CPU utilization.

51.0.6 66. Check file upload eligibility.

Hint: File size and type.

51.0.7 67. Determine storage warning level.

Hint: Disk usage percentage.

51.0.8 68. Check internet plan upgrade recommendation.

Hint: Data usage.

51.0.9 69. Determine account lock status.

Hint: Failed login attempts.

51.0.10 70. Check subscription renewal reminder.

Hint: Days remaining.

52 Industrial & Mechanical Engineering

52.0.1 71. Check machine startup permission.

Hint: Power, temperature, and safety checks.

52.0.2 72. Determine maintenance requirement.

Hint: Operating hours.

52.0.3 73. Check pressure vessel safety status.

Hint: Pressure limits.

52.0.4 74. Determine lubrication schedule.

Hint: Running hours.

52.0.5 75. Check motor overload condition.

Hint: Current exceeds rated value.

52.0.6 76. Determine fuel refill requirement.

Hint: Fuel level percentage.

52.0.7 77. Check cooling system activation.

Hint: Temperature threshold.

52.0.8 78. Determine emergency shutdown condition.

Hint: Multiple sensor readings.

52.0.9 79. Check production quality status.

Hint: Defect percentage.

52.0.10 80. Determine machine efficiency category.

Hint: Efficiency ranges.

53 Smart Systems & IoT

53.0.1 81. Smart home light automation.

Hint: Time and occupancy.

53.0.2 82. Smart irrigation activation.

Hint: Soil moisture level.

53.0.3 83. Smart AC control.

Hint: Room temperature.

53.0.4 84. Smart security alert.

Hint: Motion detection.

53.0.5 85. Smart energy-saving mode.

Hint: Power consumption.

53.0.6 86. Smart water tank monitoring.

Hint: Water level.

53.0.7 87. Smart streetlight operation.

Hint: Ambient light level.

53.0.8 88. Smart parking slot availability.

Hint: Sensor status.

53.0.9 89. Smart fire alarm activation.

Hint: Smoke and temperature.

53.0.10 90. Smart elevator floor selection validation.

Hint: Valid floor range.

54 Master-Level Scenario Problems

54.0.1 91. University admission system.

Hint: Marks + entrance exam + category.

54.0.2 92. Airline ticket pricing engine.

Hint: Class + season + age.

54.0.3 93. Hospital triage system.

Hint: Severity + age + symptoms.

54.0.4 94. Loan approval workflow.

Hint: Income + credit score + collateral.

54.0.5 95. Industrial machine control system.

Hint: Multiple safety conditions.

54.0.6 96. Employee performance appraisal system.

Hint: KPIs and attendance.

54.0.7 97. Online shopping checkout system.

Hint: Stock + payment + shipping.

54.0.8 98. ATM transaction system.

Hint: Authentication + balance.

54.0.9 99. Smart city traffic control system.

Hint: Traffic density.

54.0.10 100. Multi-level user authentication system.

Hint: Password + OTP + security question.

These **100 real-life conditional-statement problems** are excellent for practicing all forms of decision-making in C:

- `if`
- `if-else`
- `else-if` ladder
- Nested `if`
- `switch-case`
- Mixed decision structures used in real applications.

55 C Programming – Comprehensive Conditional Statements Practice Set

Objective: These problems are designed so that you are **forced to use multiple conditional structures together**:

`if if-else else-if ladder (if → else if → else) Standalone else-if chains`
`Nested if Nested if-else Nested if → else-if → else switch-case break`

Most problems require **more than one conditional construct**.

56 Level 1 – Foundation Scenarios

56.0.1 1. Student Result Management System

Hint:

- `if` → Check attendance eligibility.
 - Nested `if-else` → Pass/Fail.
 - `else-if` ladder → Grade assignment.
-

56.0.2 2. Voting and Election Eligibility

Hint:

- `if` → Citizenship check.
 - Nested `if` → Age verification.
 - `switch` → Select election type.
-

56.0.3 3. ATM Transaction System

Hint:

- `if` → PIN validation.
 - `switch` → Deposit/Withdraw/Balance.
 - Nested `if-else` → Balance checks.
-

56.0.4 4. Employee Bonus Calculator

Hint:

- `if` → Minimum service years.
 - `else-if` ladder → Bonus slabs.
 - Nested `if` → Performance rating.
-

56.0.5 5. E-Commerce Discount Engine

Hint:

- `if` → Customer validity.
 - `else-if` → Purchase amount slabs.
 - Nested `if-else` → Premium membership.
-

57 Level 2 – Academic Systems

57.0.1 6. University Admission Portal

Hint:

- `if` → Entrance qualification.
 - Nested `if-else` → Category reservation.
 - `else-if` ladder → Admission status.
-

57.0.2 7. Scholarship Evaluation System

Hint:

- `if` → Minimum marks.
 - Nested `if` → Family income.
 - `else-if ladder` → Scholarship amount.
-

57.0.3 8. Placement Eligibility Checker

Hint:

- `if` → No active backlogs.
 - Nested `if-else` → CGPA validation.
 - `switch` → Company category.
-

57.0.4 9. Online Examination Portal

Hint:

- `if` → Registration check.
 - Nested `if` → Fee payment.
 - `switch` → Subject selection.
-

57.0.5 10. Hostel Allocation System

Hint:

- `if` → Admission confirmation.
 - `else-if ladder` → Distance slabs.
 - Nested `if-else` → Room type.
-

58 Level 3 – Banking & Finance

58.0.1 11. Loan Approval Workflow

Hint:

- `if` → Income eligibility.
 - Nested `if-else` → Credit score.
 - `else-if ladder` → Interest rate.
-

58.0.2 12. Credit Card Approval System

Hint:

- `if` → Employment status.
 - Nested `if` → Salary.
 - `switch` → Card type.
-

58.0.3 13. Income Tax Calculator

Hint:

- `else-if ladder` → Tax slabs.
 - Nested `if` → Senior citizen benefit.
-

58.0.4 14. Insurance Premium Calculator

Hint:

- `if` → Medical eligibility.
 - `else-if ladder` → Age groups.
 - Nested `if-else` → Smoking status.
-

58.0.5 15. Mutual Fund Recommendation

Hint:

- `if` → Investment amount.
 - `switch` → Risk profile.
 - Nested `if` → Age category.
-

59 Level 4 – Industrial / Mechanical Engineering

59.0.1 16. Machine Startup Authorization

Hint:

- `if` → Power supply.
 - Nested `if` → Temperature.
 - Nested `if-else` → Pressure.
 - `switch` → Operation mode.
-

59.0.2 17. Boiler Safety Monitoring

Hint:

- `else-if ladder` → Temperature ranges.
 - Nested `if` → Pressure validation.
-

59.0.3 18. Predictive Maintenance Alert System

Hint:

- `if` → Operating hours.
 - Nested `if-else` → Vibration level.
 - `switch` → Machine type.
-

59.0.4 19. Production Quality Inspection

Hint:

- `if` → Defect detection.
 - `else-if ladder` → Quality grade.
 - Nested `if` → Rework decision.
-

59.0.5 20. Fuel Consumption Analysis

Hint:

- `if` → Fuel level.
 - `else-if ladder` → Efficiency rating.
 - Nested `if-else` → Warning status.
-

60 Level 5 – Healthcare

60.0.1 21. Hospital Triage System

Hint:

- `if` → Emergency condition.
 - `else-if ladder` → Severity level.
 - Nested `if` → ICU eligibility.
-

60.0.2 22. Health Insurance Approval

Hint:

- `if` → Policy validity.
 - Nested `if` → Age.
 - `else-if` ladder → Premium category.
-

60.0.3 23. Blood Donation Eligibility

Hint:

- `if` → Age requirement.
 - Nested `if-else` → Weight validation.
-

60.0.4 24. BMI Assessment System

Hint:

- `if` → Valid input.
 - `else-if` ladder → BMI category.
 - Nested `if` → Diet recommendation.
-

60.0.5 25. Surgery Approval Workflow

Hint:

- `if` → Medical clearance.
 - Nested `if` → Insurance approval.
 - `switch` → Surgery type.
-

61 Level 6 – IT & Software Systems

61.0.1 26. Multi-Level Login System

Hint:

- `if` → Username check.
 - Nested `if` → Password.
 - Nested `if` → OTP.
-

61.0.2 27. User Access Control

Hint:

- `if` → Authentication.
- `switch` → User role.
- Nested `if-else` → Permissions.

61.0.3 28. Password Strength Analyzer

Hint:

- `if` → Length.
 - Nested `if` → Digit check.
 - Nested `if` → Special character.
-

61.0.4 29. Software License Validation

Hint:

- `if` → License key.
 - `else-if` ladder → Days remaining.
 - Nested `if` → Renewal suggestion.
-

61.0.5 30. File Upload System

Hint:

- `if` → File exists.
 - Nested `if` → File size.
 - `switch` → File type.
-

62 Level 7 – Transportation

62.0.1 31. Driving License Issuance

Hint:

- `if` → Age.
 - `switch` → Vehicle category.
 - Nested `if-else` → Learner/Permanent.
-

62.0.2 32. Railway Reservation System

Hint:

- `if` → Seat availability.
 - `switch` → Coach type.
 - Nested `if` → Concession.
-

62.0.3 33. Airport Check-in System

Hint:

- `if` → Ticket validation.
 - Nested `if` → Passport.
 - Nested `if` → Visa.
-

62.0.4 34. Toll Collection System

Hint:

- `switch` → Vehicle type.
 - Nested `if-else` → FastTag.
-

62.0.5 35. Vehicle Service Reminder

Hint:

- `if` → Mileage.
 - `else-if` ladder → Service category.
-

63 Level 8 – Advanced Integrated Systems

63.0.1 36. Smart Home Controller

Hint:

- `switch` → Device selection.
 - Nested `if` → Sensor status.
 - `else-if` ladder → Operating modes.
-

63.0.2 37. Smart Irrigation Controller

Hint:

- `if` → Soil moisture.
 - Nested `if` → Weather forecast.
 - `switch` → Watering mode.
-

63.0.3 38. Smart Traffic Signal System

Hint:

- `switch` → Signal state.

- Nested `if-else` → Traffic density.
-

63.0.4 39. Smart Parking Management

Hint:

- `if` → Slot availability.
 - Nested `if` → Vehicle type.
 - `else-if` ladder → Parking fee.
-

63.0.5 40. Smart Elevator System

Hint:

- `if` → Valid floor.
 - `switch` → Direction.
 - Nested `if` → Weight limit.
-

64 Expert Level

64.0.1 41. University ERP System

Hint: Combine all conditional structures.

64.0.2 42. Banking Core Transaction Engine

Hint: Authentication + Menu + Validation.

64.0.3 43. Industrial Plant Monitoring Dashboard

Hint: Sensors + Safety Levels + Machine Modes.

64.0.4 44. Hospital Management System

Hint: Registration + Diagnosis + Billing.

64.0.5 45. Airline Reservation Engine

Hint: Ticket + Seat + Fare Category.

64.0.6 46. Online Shopping Platform

Hint: User + Product + Payment + Delivery.

64.0.7 47. Multi-Level Security Clearance System

Hint: ID + Password + OTP + Role.

64.0.8 48. Autonomous Vehicle Decision System

Hint: Obstacle + Speed + Road Condition.

64.0.9 49. Smart City Resource Management

Hint: Energy + Water + Traffic + Alerts.

64.0.10 50. Integrated Enterprise Management System

Hint: HR + Finance + Inventory + Operations.

64.1 Conditional Constructs Coverage Table

Construct	Used In
if	1–50
if-else	1–50
else-if ladder	1–50
Nested if	1–50
Nested if-else	1–50
Nested if → else-if → else	11–50
switch-case	3, 9, 12, 15, 16, 25, 27, 30, 31, 32, 34, 36, 37, 38, 40–50
break	All switch-based problems

These 50 problems are designed to force you to practice **every conditional statement type together**, similar to what is expected in semester lab exams, coding interviews, and real-world software projects.